
UCI Software IEC Target

Command Summary

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1. Introduction

1.1. Context

The “**UCI SoftwareIEC Target**” provides an optimized interface to the emulated disk drive that sits behind the SoftwareIEC interface. It bypasses the IEC protocol completely and therefore allows higher speed data transfers.

The “UCI SoftwareIEC Target” is a target of the “Ultimate Command Interface” (UCI), and is thus accessible from the cartridge port, through some I/O registers. The document “Ultimate Command Interface – Register API” describes how commands are sent over this interface.

1.2. Purpose of this document

This document describes the commands that can be sent to this target, their expected parameters, and the data returned to the C64.

2. Commands

In the Ultimate products and the Commodore 64 Ultimate, the “Software IEC Target” is accessible through target \$05. This shall be the first byte of the command.

The following paragraphs describe each of the commands of “Software IEC Target”.

2.1. SOFTIEC_CMD_IDENTIFY (0x01)

Command format: \$05 \$01

Response: Returns the string: SOFTWARE IEC TARGET V1.0.

Status: 00, OK.

2.2. SOFTIEC_CMD_LOAD_SU (0x10)

Command format: \$05 \$10 <SEC_ADDR> <VERIFY> <ADDR_LO> <ADDR_HI> <NAME>

Description: Prepares a file for loading. It opens the file and retrieves the 2-byte start address stored in the file.

Parameters:

- <SEC_ADDR>: Secondary address (\$00 for load).
- <ADDR_LO/HI>: Override load address if SEC_ADDR is 0.

Response: Returns the 2-byte start address found in the file.

Status:

- 00, OK
- 01, FILE NOT FOUND.

2.3. SOFTIEC_CMD_LOAD_EX (0x11)

Command format: \$05 \$11 <SEC_ADDR> <VERIFY>

Description: Executes the actual data transfer from the file opened by LOAD_SU directly into C64 RAM using a DMA transfer.

Parameters: If <VERIFY> is \$01, the Ultimate compares the file data against C64 memory instead of writing it.

Response: None

Status: The first byte is the status byte: \$00 for success, \$80 when a verify failure occurs. When loading, the second and third byte contain the end address of the load in little endian format.

2.4. SOFTIEC_CMD_SAVE (0x12)

Command format: \$05 \$12 <VERIFY> <SEC_ADDR> <START_LO/HI> <END_LO/HI> <NAME>

Description: Saves a block of C64 memory to a file.

Logic: The Ultimate creates the file, writes the 2-byte start address, and then dumps the memory range before closing.

Status: 00, OK or 02, SAVE ERROR.

2.5. SOFTIEC_CMD_OPEN (0x13)

Command format: \$05 \$13 <SEC_ADDR> \$00 <NAME>

Description: The OPEN command opens a file in the virtual file system and attaches it to a channel indicated by the secondary address.

Data Response: None

Status: None

2.6. SOFTIEC_CMD_CLOSE (0x14)

Command format: \$05 \$14 <SEC_ADDR>

Description: The CLOSE command closes the file that is associated with the channel indicated by the secondary address.

Data Response: none

Status: 00, OK

2.7. SOFTIEC_CMD_CHKIN (0x15)

Command format: \$05 \$15 <SEC_ADDR>

Description: Equates to the Kernal CHKIN call. It prepares the specified channel for reading data.

The target pre-fetches the first 32 bytes of data and places them in the UCI Data Channel for immediate retrieval. Further data is fetched via the *get_more_data* mechanism.

Usually, CHKIN is followed by consecutive calls to CHRIN to read the data from the channel. It would be highly inefficient to send a UCI command for every byte that is being read. Therefore, HyperSpeed kernal simply allows data to be prefetched with the CHKIN command, and reads from the data channel whenever the function CHRIN is called. This means that the UCI command is still 'open' when this happens. When CHRIN is called upon another channel, or any another command needs to be sent to the UCI, the pending CHKIN command is terminated first. This termination causes the exact number of bytes that was read from the response channel to be flushed from the stream, such that subsequent calls to CHKIN will prepare the next chunk of data from the file.

2.8. SOFTIEC_CMD_CHKOUT (0x16)

Command format: \$05 \$16 <SEC_ADDR> \$00 <DATA...>

Description: Equates to the Kernal CHKOUT call. It sends data to the specified channel.

- If SEC_ADDR is in the range of \$F0-\$FF, it performs an OPEN operation.
- If SEC_ADDR is in the range of \$E0-\$EF, it performs a CLOSE operation.
- Otherwise, it pushes the provided <DATA> bytes into the drive's write buffer.

2.9. 💡 Implementation Insights

Automatic Throttling: The *prepare_data* function in the CHKIN call initially fetches 32 bytes to minimize latency, but will scale up to 256 bytes per request if the C64 continues to pull more data.

DMA Integration: The load/save/verify routines check *is_dma_active()*. If the Ultimate is performing a DMA transfer, it pauses the C64 CPU to ensure data integrity.

Kernal Emulation: This target is specifically designed to allow custom Kernals or fast-loaders to avoid the IEC protocol entirely and access the SoftwareIEC directly through an efficient cartridge I/O mechanism.